SRM Institute of
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Delhi NCR,

Subject code - 21CSS303T Datascience

Assignment - 4

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Explain the concept of data wrangling and its importance in data analysis

→ Data wrangling involves cleaning, transforming and structuring raw data into usable format for analysis

Importance

Data wrangling ensures accurate reliable analysis by removing biases and error, boosting model performance in machine learning analysis saves time for insights over cleanup, raw data says unusable leading

Q24 Explain string manipulation in python with suitable code - ex?

Ans string manipulation in python involves operation like slicing, concatination, and formulating string, using built in methods

core operation

string are immutable sequence; access character via indexing

common methods:-

use upper()/lower() for case changes ("Hello"), replace for replace

ex text = " Python is fun! "
clean = text.strip().lower().replace
print(clean.split())

Q34 Explain the general techniques and programming practices used for efficient handling of large dataset in datascience.

Ans Efficient handling of large dataset in data science uses chunking vectorisation and distributed tools to manage memory speed.

programming practices

Leverage vectorised operation (Numpy/Pandas) over loops, optimize

Encoding data categorical data

One-hot encoding transforming categories into binary vectors avoiding original assumption but increase.

label encoding assigns integers
eg ("low" = 0, "medium" = 1)

- Dimentionality Binning
- Aggregation and generalization
- Other transformations

dtype (eg. int s, category) and
parallize with spark for distributed processing - stream data for real time need and profile memory.

Q4 Explain diff data transformation technique used in preprocessing

Ans Data transformation technique in preprocessing convert raw inconsistant data into clean standardized format suitable noise and incomplete type.

<u>standardization</u> -: Normalization
scale feature to a range typically using min-max scaling

$$x = \frac{x - \min(x)}{\max(x) - \min(x)}$$

this prevents features with larger range from dominating distance